FIRST SEMESTER M.Sc. DEGREE EXAMINATION, DECEMBER 2019

(CUCSS)

Physics

PHY IC 03—ELECTRODYNAMICS AND PLASMA PHYSICS

(2017 Admissions)

Time : Three Hours | Maximum : 36 Weightage

Section A

Answer all **twelve** *questions.*
Each question carries 1 *weightage.*

1. What are the electromagnetic boundary conditions ?
2. Explain equation of continuity in electrodynamics. What is its significance ?
3. Explain Brewster's angle and Brewster law.
4. Define vector and scalar potentials. What is the advantage of potential formulation of electrodynamics ?
5. Explain skin effect on the basis of electromagnetic wave propagation in conducting media. Write down an expression for skin depth.
6. Distinguish between electromagnetic wave propagation through transmission lines and wave guides
7. Draw the equivalent circuit for a transmission line. Write down its parameters
8. What are the postulates of special theory of relativity ?
9. Show that Lorentz gauge is invariant under Lorentz transformation.
10. What is electro magnetic field tensor ?
11. Define Plasma. Explain the criteria for ionised gases to be called as plasma.
12. Write down Boltzman- Vlasov equation. Explain its importance.

(12 × 1 = 12 weightage)

Section B

Answer any **two** *questions.*
Each question carries 6 *weightage.*

13. Derive Maxwell's equation on the basis of fundamental electric and magnetic phenomena.
14. Discuss the reflection and refraction characteristics of electromagnetic waves incident obliquely with parallel polarisation at a dielectric interface .Derive Fresnel equations.

15. Derive the general Lorentz transformation equations for electric and magnetic fields.

16. Derive equation of motion of plasma in the fluid approximation.

(2 × 6 = 12 weightage)

Section C

Answer any **four** *questions.*
Each question carries 3 *weightage.*

17. Calculate Poynting vector at the surface of sun if the power radiated by sun is 3.8×10^{26} W and the radius of sun is 7×10^8 m. Also calculate the average solar energy incident on earth which is at distance of 7×10^8 m from the sun ?

18. Derive non homogeneous wave equations satisfied by vector and scalar potentials under Lorentz gauge.

19. Derive TE wave components in rectangular wave guides and show that TE_{00} cannot exist in a rectangular wave guide.

20. A rectangular air filled cavity has the dimension a = 4cm, b = 3cm and d = 5cm. Determine the most dominant mode and its resonant frequency.

21. Show that the three dimensional volume element $dx\ dy\ dz$ is not Lorentz invariant where as four dimensional entity $dx\ dy\ dz\ dt$ is lorentz invariant.

22. Calculate the Larmor radius for a 10 keV electron in earth's magnetic field of 5×10^{-5} Tesla assuming parallel component of velocity is negligible.

(4 × 3 = 12 weightage)